BIPIN CHOWDARY

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PROFESSIONAL SUMMARY

AI engineer and M.S. Artificial Intelligence candidate at Florida Atlantic University with an engineering foundation in electrical systems, robotics, cloud computing, and AI/ML. Develops reproducible research prototypes spanning marine autonomy, carbon-aware machine learning, information retrieval, computer vision, and deep learning. Experienced in baseline design, dataset preparation, quantitative evaluation, software testing, and translating research questions into deployable systems.

EDUCATION

Master of Science in Artificial Intelligence | 2025 - Expected Dec 2026
Florida Atlantic University, Boca Raton, FL
Major: Artificial Intelligence | Coursework: Deep Learning, Data Mining & ML, Information Retrieval, IoT Security

Bachelor of Technology (Honors) in AI & Machine Learning | 2021 - 2024
Quantum University, Roorkee, India
Minor: Robotics & Automation | First Class with Distinction; Fourth-year topper

Diploma in Electrical & Electronics Engineering | 2018 - 2021
A.A.N.M. & V.V.R.S.R. Polytechnic, Gudlavalleru, India
Minor: Robotics | Electrical systems, Electronics, Sensors, and Automation

PROFESSIONAL EXPERIENCE

Mail & Logistics Operations Associate – Exela Technologies, Florida Atlantic University | Boca Raton, FL | Oct 2025 - Present

- Operate digital package-management workflows for parcel intake, recipient verification, automated notifications, exception tracking, and service-desk resolution.
- Coordinate campus mail-route handoffs while maintaining traceable chain-of-custody, package-status, and recipient-delivery records across Central Receiving.

Electrical Engineering Intern - APGENCO | PowerGrid | 6 months

- Increased grid reliability by 20% through optimization of electrical systems in collaboration with engineering teams.
- Contributed to maintenance projects that reduced downtime by 15% over six months.

Web Development Intern - Lets Grow More | Portfolio and Web Design | 2 months

- Improved client online presence by designing five user-friendly websites, resulting in a 45% increase in engagement.
- Enhanced user engagement metrics by 50% through implementation of best practices in web design.

Cloud Computing Intern - Google Cloud | DevOps and Cloud Engineering | 6 months

- Streamlined cloud deployment processes, reducing service delivery time by 40% through optimized infrastructure.
- Enhanced operational efficiency by 20% through collaborative projects focusing on process improvements.

AI & Machine Learning Intern - Xebia | AIML Algorithms and Deep Learning Architectures | 36 months

- Developed advanced AIML models, contributing to two academic research publications.
- Successfully implemented deep learning architectures, achieving 85% model accuracy on real-world applications.

AI & Robotics Automation Intern - RoboLabs AI | Automation and Robotics | 6 months

- Improved robotic function efficiency by 30% through automation solutions integrated with AI technologies.
- Led a team in automation projects, resulting in a 25% reduction in operational time.

SELECTED RESEARCH & ENGINEERING PROJECTS

AquaNavAI - Energy-Aware Marine Navigation | Independent Research Prototype | 2026 | Live Demo | Read Paper

- Built a reproducible ASV route-planning platform with A* and Dijkstra baselines, a propulsion-energy proxy, and an offline MapLibre mission visualization.
- Created 30 paired evaluation cases and implemented provenance, checksums, schemas, CI, automated tests, and NOAA-source adapters.

Carbon-Aware Marine AI | Graduate Research Project | 2026 | View Case Study | Read Paper

- Prepared NOAA OISST 2020-2026 data; achieved Ridge RMSE of 0.1336 versus 0.1451 for persistence on the South Florida study subset.
- Used CodeCarbon to estimate approximately 99.5% carbon savings under the evaluated clean-energy regional schedule.

Automotive Modding Information Retrieval | Graduate Information Retrieval Project | 2026 | View Project | Watch Walkthrough

- Constructed a 693-document corpus with 10 queries and 200 relevance judgments; compared BM25, dense retrieval, and hybrid ranking.
- BM25 reached P@10 of 0.46 and nDCG@10 of 0.3858, outperforming the tested dense and hybrid configurations.

Travel Green: Carbon-Aware Mobility and Credit Platform | Sustainable Mobility Web Platform | 2026 | View Web Repo

- Developed a carbon-aware mobility platform that helps users compare transportation options, estimate emissions, and explore incentive-based carbon credits for more sustainable travel decisions.

ADDITIONAL ENGINEERING & AI PROJECTS

Fuel Efficiency Prediction to Reduce Carbon Emissions | Machine Learning | 2024 | Live Demo
- Developed a Streamlit-based prediction application covering data preparation, model comparison, evaluation, and practical fuel-efficiency insights.

Crowd Management and Anomaly Detection | Computer Vision | 2024 | Check Project
- Built a computer-vision workflow for crowd-pattern analysis and anomaly detection, emphasizing public-safety use cases and explainable evaluation.

Navigation of a 4-Omni-Wheeled Robot | Robotics | 2022 | View Case Study
- Studied gradient-based navigation, obstacle avoidance, and predictive-control concepts for an omni-directional mobile robot.

Autonomous Smart City | Electronics & Robotics | 2019 | Watch Demo
- Integrated a pilot-less railway prototype, remote-operated construction crane, sensors, and automation logic into a physical smart-city model.

Deep Convolutional Generative Adversarial Network | Deep Learning | 2022 | View Project
- Implemented and evaluated a DCGAN image-generation pipeline, documenting training behavior, generated samples, and model limitations.

Industry-Specific Web Platforms | Software Development | 2023
- Designed responsive portfolio, industrial-product, hostel, and education-oriented web interfaces with deployment-focused front-end workflows.

TECHNICAL SKILLS

Programming & Data: Python, R, MATLAB, NumPy, Pandas, Matplotlib, Seaborn
AI / Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, deep learning, model evaluation, information retrieval
Computer Vision: OpenCV, image preprocessing, segmentation, anomaly detection
Robotics & Embedded: ROS, Gazebo, Arduino, Raspberry Pi, sensors, control logic, navigation algorithms
Cloud & DevOps: Google Cloud, Kubernetes, Docker, CI/CD, load balancing, cloud networking, infrastructure automation
Research Engineering: experiment design, baseline comparison, reproducibility, provenance, JSON Schema, CodeCarbon
Software & Deployment: Git, GitHub, TypeScript, JavaScript, Streamlit, MapLibre, Cloudflare Pages

RESEARCH MANUSCRIPTS & TECHNICAL REPORTS

- AquaNavAI: Reproducible Energy-Aware Route Planning for Autonomous Surface Vehicles - technical manuscript and software artifact, 2026.
- Carbon-Aware Marine AI: Regional Carbon-Aware Sea-Surface-Temperature Forecasting - graduate research report, 2026.
- Automotive Modding Retrieval: Empirical Comparison of BM25, Dense, and Hybrid Retrieval - evaluation report, 2026.
- Multi-Task Radiotherapy Planning: Integrated Segmentation and Dose Prediction - deep-learning project report, 2025.

Full project pages and available papers: Open portfolio and papers

AWARDS & RECOGNITION

• Innovation Award – National Science Fair Recognized for developing a cutting-edge robotic automation model with a focus on sustainable solutions.
• Engineering Design Excellence – Google Awarded for innovative contributions to AI-powered engineering systems during a collaborative design project.
• Cloud Engineering Certified – Google Cloud Achieved for outstanding performance in advanced cloud solutions and infrastructure optimization.
• DevOps Essentials Certified – Google Recognized for mastery in DevOps practices, significantly improving deployment pipelines.
• Cloud Architecture Specialist – Google Cloud Certified for exceptional skills in designing and deploying scalable cloud architectures.
• Web Development Honor – CSS Winner Award Received for designing an industry-leading, visually stunning, and highly functional website.
• Generative AI & LLM Innovator – NVIDIA Honoured for pioneering applications of Generative AI using NVIDIA's frameworks and tool.
• Academic distinction: B.Tech First Class with Distinction; fourth-year class topper.

GOOGLE CLOUD CREDENTIALS

Google Skills profile: Silver League | 20,200 points | 15 earned badges | Member since 2022 | View public profile
- Implement DevOps Workflows in Google Cloud | Develop Your Google Cloud Network
- Build a Secure Google Cloud Network | Set Up an App Dev Environment on Google Cloud
- Implementing Cloud Load Balancing for Compute Engine | Cloud Engineering
- DevOps Essentials | Kubernetes in Google Cloud
- Automate Deployment and Manage Traffic on a Google Cloud Network | Understand Your Google Cloud Costs
- Baseline: Infrastructure | Google Cloud Essentials

RESEARCH INTERESTS

- **Marine AI & Ocean Engineering:** Intelligent navigation, environmental monitoring, ocean-data analysis, and decision-support systems for marine applications.
- **Autonomous Systems & Robotics:** Perception, route planning, obstacle avoidance, control, and energy-efficient operation of autonomous vehicles and robotic platforms.
- **Carbon-Aware Computing:** Energy-efficient machine learning, emissions-aware cloud deployment, regional workload scheduling, and sustainable AI experimentation.
- **Computer Vision:** Image preprocessing, segmentation, object and anomaly detection, environmental perception, and vision-assisted automation.
- **Information Retrieval:** BM25, dense and hybrid retrieval, semantic search, relevance evaluation, and domain-specific knowledge systems.
- **Research Software Engineering:** Reproducible pipelines, data provenance, automated testing, CI/CD, experiment tracking, and deployment-ready research prototypes.
- **Environmental Data Intelligence:** Machine-learning analysis of oceanographic, climate, sensor, and geospatial datasets for forecasting and operational planning.
- **Patent-Oriented Applied Innovation:** Converting interdisciplinary AI, robotics, electrical-engineering, and marine-research concepts into practical systems and defensible intellectual property.